



COM 2301 PROJECT PART B

CAMPUS PROBLEM-SOLVING THROUGH FEASIBILITY STUDY AND PROTOTYPE DEVELOPMENT

DUE DATE: 23/09/2026

PRESENTATION DATE: 29-30 September 2026

ASSIGNMENT OVERVIEW:

Universities experience various challenges that affect students, staff, academic activities, administration, communication, safety, access to services, and the overall campus experience.

Many of these challenges can potentially be addressed or improved through innovative technological solutions.

In this assignment, you are required to identify a real problem that you have observed or experienced on campus and investigate whether developing a technological solution to address the problem would be feasible.

You will then design and develop a prototype demonstrating how your proposed solution could work.

The purpose of this assignment is not simply to develop an application. You must first demonstrate that: **a real problem exists, the problem is worth solving, your proposed solution is appropriate, and the solution is feasible within the identified environment.**

Task

Working in groups, identify one problem you have observed on campus and propose a technological solution to address it.

You are required to:

1. Identify and describe the campus problem.
2. Propose a technological solution.
3. Conduct a feasibility study.
4. Design the proposed system.
5. Develop a functional prototype.
6. Present and demonstrate your solution.

1. Problem Identification (10 Marks)

Describe:

- The problem you identified.
- Where the problem occurs.
- Who is affected.
- How the current process works.
- Why the problem needs a solution.

The problem must be based on your own observations or publicly available information.

2. Proposed Solution (10 Marks)

Describe your proposed technological solution.

Explain:

- What the system will do.
- Who will use it.
- The main features.
- How it will address the identified problem.

3. Feasibility Study (25 Marks)

Evaluate the proposed solution in terms of:

- **Technical feasibility:** Technology and resources required.
- **Economic feasibility:** Estimated development and operational costs.
- **Operational feasibility:** How the solution would work within the campus environment.
- **Schedule feasibility:** Estimated development time.

Conclude whether the solution is:

Feasible / Feasible with Conditions / Not Feasible

4. System Design (15 Marks)

Provide:

- Functional requirements
- Non-functional requirements
- Use Case Diagram
- Database/System Design
- User-interface designs

5. Prototype (25 Marks)

Develop a **functional prototype** demonstrating the main features of your proposed solution.

The prototype must be interactive. Static screenshots alone are not sufficient.

Use **fictional/sample data** in the prototype.

6. Testing (5 Marks)

Test the main functions of the prototype and provide:

Test → Expected Result → Actual Result → Pass/Fail

7. Presentation (10 Marks)

Present and demonstrate:

Problem → Proposed Solution → Feasibility → Design → Working Prototype

Submission

Submit:

- **Project Report (PDF TO MOODLE)**
- **Source Code (TO GITHUB)**
- **Presentation (PROTOTYPE TOGETHER WITH SLIDES)**